

Tarun Mittal

Senior Software Engineer

(+91) 9079727197 | tmittaliet@gmail.com | linkedin.com/in/t-mittal | github.com/TM-DEV-PRO | Bangalore, India

SUMMARY

- Senior Software Engineer with **5 years** owning cloud-native, high-throughput distributed systems end to end.
- Deep in **Java and Spring Boot microservices**, Hibernate/JPA, event-driven architectures, plus **Agentic AI and RAG**.
- Track record shipping production backends, leading migrations, and improving reliability, latency, and scale.

SKILLS

- Languages** – Java, SQL, Python, C, C++.
- Backend & APIs** – Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, REST, gRPC, Spring Batch, Design Patterns, Service-Oriented Architecture (SOA).
- AI & Agents** – Python (FastAPI, LangGraph, LangChain), LLM agents (OpenAI/Gemini/Claude), MCP, RAG, Milvus, pgvector.
- Streaming & Data** – Apache Kafka, Apache Flink, ClickHouse, BigQuery, Batch Processing.
- Databases & Storage** – PostgreSQL, MySQL, Redis, MongoDB, Elasticsearch, Amazon S3, Caching.
- Cloud & Infrastructure** – GCP, AWS, Docker, Kubernetes, CI/CD, Maven, Bazel, ELK, Grafana, New Relic, Datadog, PostHog, LangSmith.
- Core Engineering** – Distributed Systems, HLD/LLD, Multithreading, Concurrency, Sharding, Mentorship, JUnit, Mockito.

PROFESSIONAL EXPERIENCE

Senior Software Engineer IMPACT ANALYTICS, Bangalore May 2026 – Present

Agentic AssortSmart (AI-Powered Retail Merchandise Planning)

- Building **AssortSmart** so retailers decide **what to buy, how much, and for which stores** via a planner copilot (**Python, LangGraph, MCP**) on a **Spring Boot** doing layer, targeting plan turnaround from days toward **under 1 hour** and search from **1 to ≥ 20 configs/plan**.
- Cutting measured clustering-run failures from **8.5% (37/437 on kik)** toward **under 2%** (target) with **14** audited read-only tools and **3** human confirm gates so the agent never writes SQL.
- Replacing a projected **12B-row** store-week design with a **$\sim 25M$ -row** aggregate so month rollups stay **sub-second** and cell edits stay **~ 0.4 ms** (measured) without materializing the explosion.
- Adopting per-tenant **ClickHouse (63 tables / 8 layers, insert-only / partition-swapped, agent read-only)** as the agentic planning store after a row-identical POC cut **250M-row** pivots from **189s to 12.3s** ($\sim 15.5x$, measured).
- Tech** – Java, Spring Boot, Python, FastAPI, LangGraph, MCP, ClickHouse, BigQuery, GCS, Datadog, LangSmith, PostHog, Docker.

Software Development Engineer 2 UBER (via EPAM Systems), Bangalore July 2024 – May 2026

Financial Risk Management (FRM) Scoping Platform (Uber Finance)

- Owned design and architecture for Uber's **FRM Risk Scoping platform (Spring Boot, MySQL, React)** across **8 screens** and **30+ REST APIs**, automating quarterly scoping at **\$340M group materiality** and targeting a **70%** cut in manual reconciliation (**~ 2 weeks to $\sim 3-4$ days**).
- Designed an **11-table MySQL** schema on Spring Data JPA/Hibernate with controller, service, and repository layers that auto-flags material items across **55 line items** and **14 entities**.
- Owned Sheets to **MySQL v2 recon migration** across **18 files**, validating HFM extracts against public **10-Q** filings and fixing a latent column-aliasing bug.
- Led 3 engineers** through design reviews, API contracts, and CI quality gates for releases feeding PwC audit work papers.
- Tech** – Java, Spring Boot, Spring Data JPA, Hibernate, MySQL, JUnit, Docker.

Menu Ingestion & Automation Platform (Uber Eats)

- Built an event-driven menu pipeline (**Selenium $\rightarrow$ Kafka $\rightarrow$ Flink**) that brought onboarding from **24h to 2h** and saved **\$600K+ annually** on **30K+ menus/month**.
- Built a **LangChain RAG + Gemini 2.5 Pro** extractor over **Milvus** that turns multilingual **PDF/image** menus into Uber Eats schema at **98% fidelity / 100% schema consistency** (offline eval).
- Raised successful ingestions to **95%+** with IP rotation, dynamic proxy pools, and adaptive backoff against anti-bot defenses.
- Tech** – Python, Selenium, Kafka, Flink, LangChain, Gemini, RAG, Milvus, GCP, Docker.

ANZ Driver Document Compliance (Uber Mobility)

- Automated Uber driver and vehicle document checks for the **ANZ** region, achieving **99.9%** compliance with local authorities and eliminating about **20 hours/week** of manual verification.

Software Development Engineer 2 MASTERS INDIA, Noida December 2022 – June 2024

GST Compliance & E-Invoicing Platform

- Designed and operated a **PHP (Laravel)** to **Spring Boot microservices** strangler for **1,500+ clients**, bringing p95 from **1.2s to 300ms** while mentoring **2 engineers**.
- Built an event-driven IRP path on **Kafka** and **quarter-sharded PostgreSQL** that processed **1M+ submissions/day** and **100K+/import** and lifted throughput from **700 to 4,000 requests/min**.
- Hardened safe IRP replay with **idempotent sinks (idempotency keys, retries, DLQ)** and Redis cache-aside that lowered redundant DB reads **30%**.
- Established **ELK / New Relic** on-call alerting and a client usage dashboard that lowered incident triage **70%** and support tickets **35%**, raising coverage **35% to 82%** at **98%** deploy success.
- Tech** – Java, Spring Boot, Hibernate, Kafka, PostgreSQL, MongoDB, Redis, Elasticsearch, Spring Batch, Docker, ELK, New Relic, AWS.

Software Development Engineer GEEKSFORGEEKS, Noida August 2021 – November 2022

- Stabilized doubt-support for **10K+ daily queries** and **10x** contest spikes by migrating the backend from **PHP to Spring Boot (Java)**.
- Lifted premium subscriptions **15–20%** by shipping voting, pinning, and locking REST APIs on **MySQL, MongoDB, Redis, and Elasticsearch**.
- Raised course sales **30%** by launching an influencer earnings and analytics dashboard (transactions, coupons, filters).
- Raised operational efficiency **70%** by orchestrating cron pipelines for video processing, reminders, and recording cleanup.

EDUCATION

B.Tech (Information Technology) Institute of Engineering & Technology, Lucknow July 2017 – June 2021

ACHIEVEMENTS & CERTIFICATIONS

- Ranked **2260 / 37,000+** in the Google Code Jam Qualifier 2021 and finalist (top 3 nationally) at the **Smart India Hackathon 2020 (MHRD)**.
- Participated in the **Global AI Hackathon** at EPAM Systems.
- Earned **HackerRank Problem Solving** and **LangChain Academy** certifications (**HackerRank, LangChain**).